

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1809

Somerville 345 kV
May 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1809. The Interconnection Customer proposes a 343.8 MW natural gas combined cycle facility interconnecting at the Somerville 345 kV substation in the Pennsylvania area. The request was received on July 07, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 456 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,071,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Somerville 138 kV/1 kV Transformer Bank	Burwell – P.T.S Substation 345.0 kV	111.8	424	24.3
Thermas – Kilarc 138.0 kV Line	Thermas – Conemaugh Hydro Plant 138.0 kV	104.4	177	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Somerville 138 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$6,558,000	94% (\$6,146,000)
Thermas – Kilarc 138.0 kV Line	Reconductor line	\$1,925,000	100% (\$1,925,000)
	Total allocated to project		\$8,071,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.